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ALPHA CHOICE INNOVATIVE ACADEMY

(International Secondary School)

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BASIC TECHNOLOGY

THIRD TERM 2025/2026 SESSION

GRADE 8

SYNOPSIS

WEEEEK

TOPIC

1. REVISION OF SECOND TERM WORK
2. BELTS AND CHAIN DRIVES I
3. BELTS AND CHAIN DRIVES II
4. GEARS I
5. GEARS II
6. HYDRAULIC AND PNEUMATIC MACHINES
7. MID - TERM BREAK
8. BUILDING CONSTRUCTION I (SITE PREPARATION)
9. BUILDING CONSTRUCTION II (SETTING OUT)
10. REVISION
11. EXAMINATION
12. EXAMINATION

WEEK ONE: REVISION

WEEK: TWO: BELT AND CHAIN

DRIVES I

OBJECTIVES:

1. Identify belt and chain drives
2. Highlight properties of belt and chain drives
3. Outline the uses of belt and chain drives

BELT AND CHAIN DRIVES

Belts are made of rubber and materials whose length and nature change with temperature. Examples of belts are motor fan belts, grinding machine belts, sewing machines. etc.



Chain and Belt drives

Chains are made of metals. Common examples are bicycle and motorcycle chains.

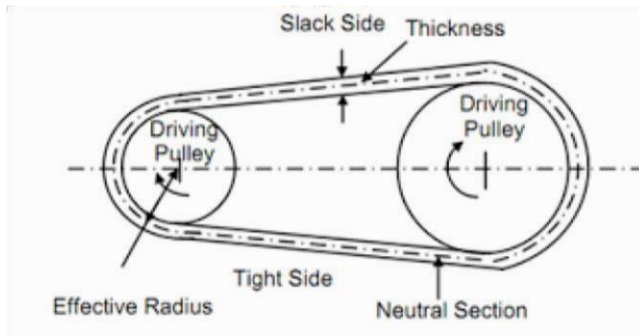
Friction is greatly needed in belt and chain drives. Belts and chains cannot work when friction is absent. Belts and chains are used in pulley mechanisms.

Belt and chain drives are used to:

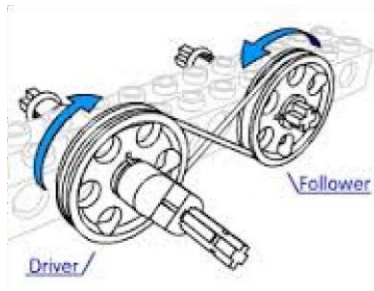
1. Transmit energy from one shaft to another when both shafts are separated by a distance greater than that require for gears.
2. Change the speed of a pulley.
3. Change the running direction of a pulley.
4. Carry materials as conveyors from one point to another.

TYPES OF BELT AND PULLEYS

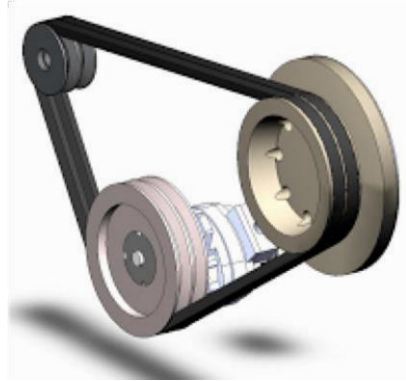
1. Open belt



2. Crossed belt



3. V-belt



4. Chain belt.

ASSIGNMENT

1. Differentiate between belt and chain drives
2. Highlight 5 uses of belt and chain drives
3. Describe with the aid of suitable diagram, belt and chain drives

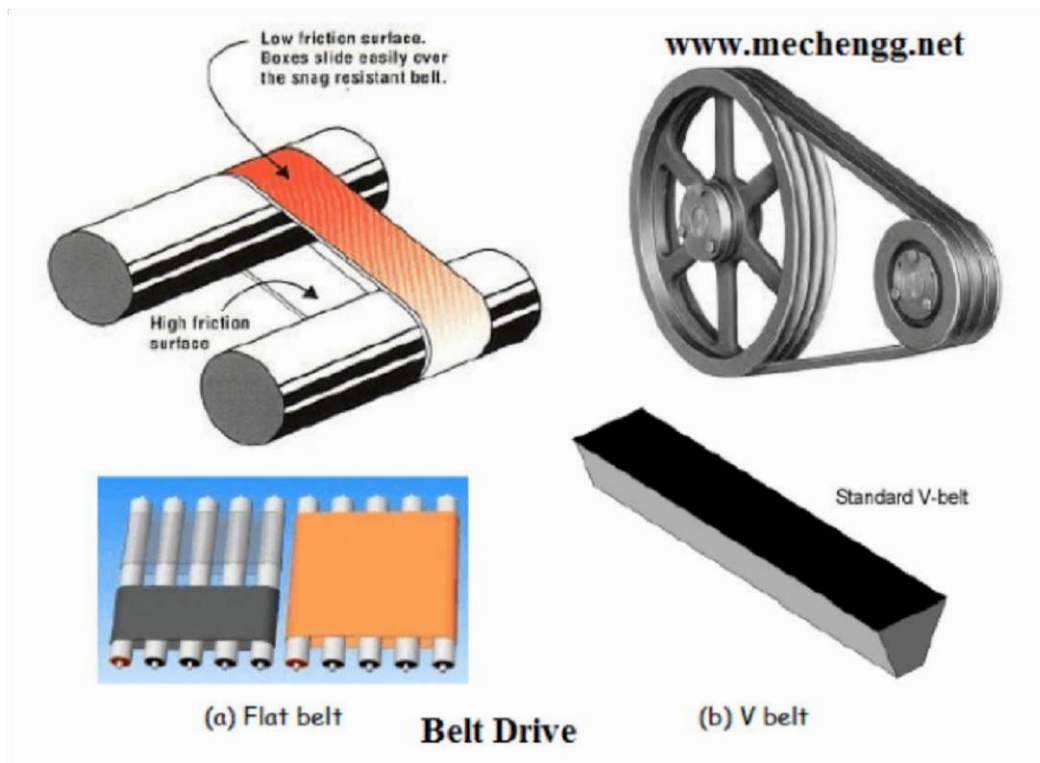
WEEK THREE: BELT AND CHAIN DRIVES II

OBJECTIVES:

1. Identify the types of belt drives
2. Highlight the differences between chain and belt drives.
3. Outline advantages and disadvantages each of belt drives.
4. Describe the term 'chain drives'.

BELT DRIVE

To transmit power from one shaft to another, pulleys are mounted on the two shafts. The pulleys are then connected by an endless belt passing over the pulleys. The connecting belt is kept in tension so that motion of one pulley is transferred to the other without slip. The speed of the driven shaft can be varied by varying the diameter of the two pulleys.



ADVANTAGES BELT DRIVES

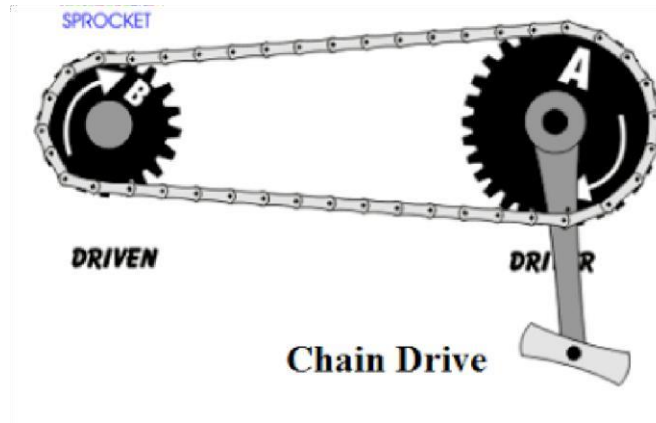
1. Made from flexible material, it can be used for the shafts whose axes are not parallel.
2. Lubricant is not required.
3. Noise is relatively less.
4. It is flexible, it slips or break during overload and safeguards the machine.
5. Initial cost is low.

DISADVANTAGES OF BELT DRIVES

1. Drive is not positive as the belt slip over the pulleys.
2. Occupies relatively more space.
3. Speed ratio cannot be maintained because of slipping of belt.
4. Tension adjustment in the belt is required from time to time.
5. Its life span is relatively low.

CHAIN DRIVES

The chain drive consist of three elements – driving sprocket, driven sprocket and endless chain wrapped around the sprocket as shown in fig. Pin joint contains, pin, bush and roller to minimize the friction and such chains are known as roller bush chain. The chain drive is positive drive where there is no slip & constant velocity ratio can be maintained. Chain drive used in bicycle, motorbike, printing machine, textile machine etc.



ADVANTAGES CHAIN DRIVES

1. Positive drive as there is no slip, hence constant velocity ratio.
2. Occupies less space compared to belt drive.
3. Its life span is more compared to belt drive.
4. Used for large center distance.
5. Transmission efficiency is larger than belt drive.

DISADVANTAGES OF CHAIN DRIVES

1. Noisy compared to belt drive.
2. Initial cost is higher compared to belt drive
3. Adjustment in the center distance is necessary.
4. Maintenance cost is higher & complex compared to belt drive.

ASSIGNMENT

1. _____ can be defined as the movement of mechanical components in engineering systems.
 - A. motion
 - B. friction
 - C. fiction

D. rotary motion

E. linear motion

2. There are_____ types of motion

A. 3

B. 4

C. 5

D. 6

E. 2

3. _____ is a type of motion that occurs in form of a straight line.

A. linear motion

B. rotary motion

C. motion

D. friction

E. fiction

4. _____ is the type of motion that occurs in circular form either clockwise or anti clockwise direction

A. fiction

B. friction

C. rotary motion

D. linear motion

E. motion

5. _____ is a device used for transmitting intermittent motion in one direction only

A. clutch

B. gear

C. brakes

D. engine

E. ratchet

6. _____ is a machine use for lifting loads?

A. pulleys

B. screw jack

C. bench vice

D. brakes

E. engine

7. _____ is a wheel with a grooved rim in which a rope or chain passes

A. pulleys

B. bench vice

C. brakes

D. wheel barrow

E. engine

8. Highlight FIVE advantages and disadvantages each of belt drives.
9. Describe the effect of friction on belt and chain drives.
10. Differentiate between belt and chain drives.

WEEK FOUR: GEARS I

OBJECTIVES:

1. Identify gears.
2. Highlight all the types of gears.
3. Outline all the uses of gears.
4. Describe the materials used in the production of gears.

GEARS: DEFINITION

Gears are Power transmission elements. It is the Gears that decides the torque, speed and direction of rotation of all the driven machine elements. Broadly speaking, Gear types may be grouped into five major categories. They are Spur, Helical, Bevel, Hypoid, and Worm. A lot of intricacies are there in the different types of gears. Actually The choice of gear type is not a very easy process. It is dependent on a number of considerations. Factors that go into it are physical space and shaft arrangement, gear ratio, load, accuracy and quality level.

A number of gears are manufactured using different materials and with different performance specifications depending on the industrial application. These gears are available in a range of capacities, sizes and

TYPES OF GEARS

1. SPUR GEAR



2. HELICAL GEAR



3. HERRINGBONE GEAR



4. RACK AND PINION



5. BEVEL GEAR



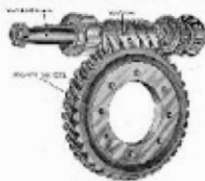
6. SPIRAL BEVEL GEAR



7. SCREW GEAR



8. WORM & WORM WHEEL



9. MITER GEAR



10. INTERNAL GEAR



speed ratios, but the main function is to convert the input of a prime mover into an output with high torque and low RPM. These range of gears find use in almost every industry right from agriculture to aerospace, from mining to paper and pulp industry. Some of the popular types of gears in use are:

Spur Gears



Spur Gear

Spur gears are straight-toothed gears having radial teeth used to transmit power and motion between parallel axes. These gears are widely used for speed increase or reduction, high torque, resolution for positioning systems. These gears can either be mounted on a hub or a shaft. The gears are available in different size, design, shape and also offer a variety of features and functions to cater to different industrial requirements.

Materials Used

Spur gears are fabricated from superior quality materials, like:

- Metal- steel, cast iron, brass, bronze and stainless steel.
- Plastic- acetal, nylon and polycarbonate.

Materials used to manufacture these gears are used keeping in mind certain factors including design life, power transmission requirements, noise generation.

Important Specifications to be Considered

- Gear center

- Bore diameter
- Shaft diameter

Use of Spur Gears

These gears find wide application in a number of fields including :

- Automobiles
- Textiles
- Industrial engineering

Bevel Gears



Bevel Gears

Bevel gears are mechanical devices used for transmitting mechanical power and motion. These gears are widely used for transmitting power and motion between nonparallel axes and are designed to transmit motion between intersecting axes, generally at right angles. The teeth on bevel gear can be straight, spiral or hypoid. The gears are suitable when the direction of a shaft's rotation needs to be changed. **Materials used** Materials used to manufacture these gears are used keeping in

mind certain factors including design life, power transmission requirements, noise generation. Some of the important materials used are :

- Metal – Steel, cast iron and stainless steel.
- Plastic – Acetal and polycarbonate.

Important specifications to be considered

- Gear center
- Bore diameter
- Shaft diameter

Use of Bevel Gear

These gears find wide application in a number of fields including :

- Automotive industry
- Textile industry
- Industrial engineering products

Helical Gears



Helical Gear

Helical gear is a popular type of gear having its teeth cut at an angle, thus allowing for more gradual and smoother meshing between gear wheels. The helical gears are a refinement over spur gears. The teeth on helical gears are specially cut at an angle, so as to face the gear. As two teeth on the gear system engage, it starts a contact on one end of the tooth which gradually spreads with the gear rotation, until the time when both the tooth are fully engaged.

The gears are available in different sizes, shapes and designs to meet the customer specifications.

Materials Used

These gears can be manufactured from superior quality materials including stainless steel, steel, cast iron, brass etc. depending on the application. **Use of Helical Gears** These gears are used in areas requiring high speeds, large power transmission, or where noise prevention is important.

- Automobiles
- Textile
- Aerospace

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- Conveyors
- Worm Gears**



Worm Gear

A worm gear is a type of gear, engaging with a worm to significantly reduce rotational speed, or allowing higher torque to be transmitted. The gear can achieve a higher gear ratio than spur gears of the same size. **Materials Used**

Worm gears can be constructed from a number of materials depending on the end application. Some of the popularly use materials are :

- Brass
- Stainless steel
- Cast iron
- Aluminum
- Hardened steel

The gears can operate under difficult conditions and have the ability to achieve large speed reductions. The gears also transmit high loads at high speed ratios.

ASSIGNMENT

1. Describe the term “gears”
2. Highlight 5 types of gears

WEEK FIVE: GEARS II

OBJECTIVES:

1. Identify gears.
2. Highlight all the types of gears.
3. Outline all the uses of gears.
4. Describe the materials used in the production of gears.

Types of Worm Gears

- Non-throated
- Single-throated
- Double-throated

Use of Worm Gears

These gears find application in :

- Electric motors
- Automotive components

Ground Gears



Ground Gears

As generally seen grinding is most of the time conceived in context of quantity fabrication of superior quality gears as a form of secondary

refining procedure. We incline to forget that grinding is essentially a basic process in the step towards production of case hardened gears. Moreover, the teeth of precision-engineered fine-pitch gears completely ground from the blank itself. The advent of trawling also led to the development and manufacturing of Ground Gears. Ever since then ground gears have made substantial improvement in the terms of designing and component accuracy. These gears assure high transmission accuracy and deliver superior efficiency, greater load capacity, and correction of profile and durability.

Ground gears can be made using different materials, such as cast iron, carbon steel, alloy steel, hardened steel, bronze, and more.

Advantages of Ground Gears

Ground Gears offer various advantages to its users, some of which are:

- **High Precision:** Achieving high precision is not a difficult task for ground gears since in the grinding process, there is little removal of material in the final pass.
- **Superior Surface Finish:** Grinding makes the surface of ground gears more shiny than that obtained from any other machining technique.
- **Improved Flexibility:** Hardened steel alloys can be used to develop into ground gears that give it added flexibility.
- **Minimal Surface Stress:** There is minimum residual surface stress in ground gears.
- **Load Carrying Capacity:** Ground Gears exhibit a higher load carrying capacity.

- **Minimal Wear and Tear:** Ground gears have minimal wear and tear that results in prolonged life.

Limitations of Ground Gears

Though ground gears offer multiple benefits and advantages, they too have some limitations:

- There is a limit to grinding procedures and that is to ferrous material.
- Hard metals can be grind in an efficient and better way than the soft ones.
- In case of worm or helical gears, grinding may not be the ideal solution. This is due to the reason that it often involves deviations in terms of removal and profile.
- Gear grinding machines are not as popular as hobbing machinery.
- Grinding demands higher costs, as it is a secondary operation.

Applications

These gears find wide application in a number of fields including :

- **High Speed Rotation:** Ground gears are ideal for uses in applications that need noise and vibration resistance in the case of high-speed gear drives. An example can be that of ground spur gears.
- **Positioning:** CP Racks and Pinions are recommended for perfect positioning applications. In these cases, ground gears are used in calculating for reducing pitch errors.

Precision Gears



Precision Gear

Precision gears are custom-made actuators that can be designed for varying uses and applications. These gears are generally used in applications under conditions of light loading. Precision gears are generally preferred for their precise, smooth, compact, noiseless and reliable performance. Precision gears can be manufactured as per the customer's drawings or based on a functional description depending on the type of application. The different types of precision gear products include, – spur gears, helical gears, worm gears, anti-backlash gears, cluster gears, clutch gears, face gears, planetary gears, gear assemblies, gear boxes, bevel gears, miter gears, metric gears, internal gears, idler gears, gear rack & pinion, worms, worm shafts, splines, spline shafts, se shafts, and more.

These gears can be manufactured as per the exact customer specifications or according to application need.

The quality and performance of a precision gear depends on the quality of blank in which it is cut. Thus it is essential to hold tight tolerances without grinding.

Precision gears are known for their trouble free superior performance, long service life, and excellent surface finish and customization capability. These gears are used in a variety of industrial applications, such as heavy machinery Industry, metal casting, metal processing, construction, and more.

Rack Gears



Rack and Pinion Gears

A rack is generally used for converting rotational motion into linear motion. It is a flat bar onto which the teeth of a pinion gear are engaged. It is a kind of gear whose axis is at infinity. These gears are designed to accommodate a wide variety of applications.**Materials Used**

A variety of materials are used keeping in mind the application. Some popularly used materials are :

- Plastic
- Brass
- Steel
- Cast Iron

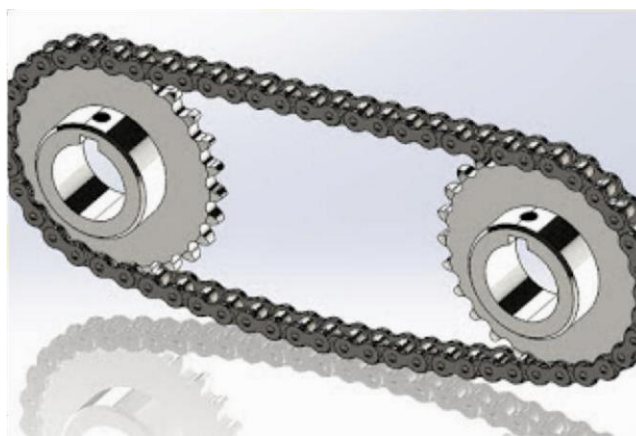
These gears ensure quieter and smoother operation. The mechanism provides less backlash and greater steering feel.

Use of Rack Gear

The gear is commonly used in steering mechanism of cars. Other important applications of rack gears include :

- Construction equipment
- Machine tools
- Conveyors
- Material handling
- Roller feeds

Sprockets



Sprocket wheel

A sprocket is a gear having metal teeth that meshes with a chain. Also known as a cog wheel, it is a small toothed ring that can fit onto the rear

wheel. It is a thin wheel having teeth that engage with a chain.**Materials Used**

A variety of materials can be used to manufacture superior quality sprockets used in different industries. Some of the materials used are :

- Stainless steel
- Hardened steel
- Cast iron
- Brass

Use of Sprockets

This simple gear finds application in diverse areas including :

- Food industry
- Bicycles
- Motorcycles
- Cars
- Tanks
- Industrial machines
- Movie projectors and cameras

Segment Gears



Segment Gears

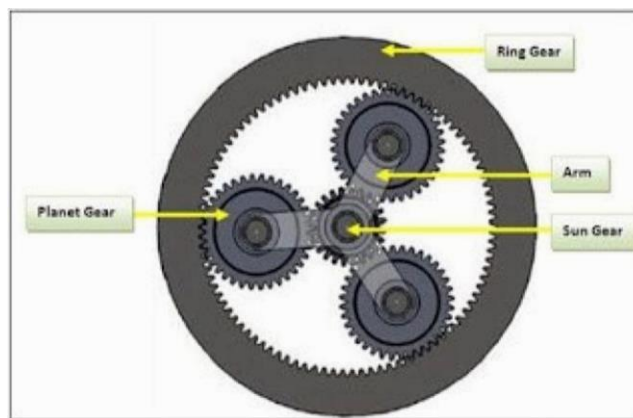
The segment gear, as the name suggests, is basically a gear wheel. These gear wheels are composed of a large number of pieces that are small parts of a circle. A segment gear is connected to the arms or trappings of the water wheel. The segment gear comes with a part for receiving or communicating the reciprocating motion from or to a cogwheel. These gears also comprise of a sector of a circular ring or gear. There are also cogs on the periphery.

Segment Gears are available in various finishes, such as untreated or heat-treated and can be designed as a single component or as an entire system.

Applications

Segment gears, which are basically gear wheels, are used in a variety of industrial uses and applications. These gears offer various advantages such as improved flexibility, superior surface finish, high precision and minimum wear and tear. Some of the uses of segment gears include:

- Defense
- Rubber
- Railways Planetary Gear



Planetary Gears

Planetary gear is an outer gear that revolves around a central sun gear. Planetary gears can produce different gear ratios depending on which gear is used as the input, which one is used as the output. **Materials**

Used The gears can be constructed from a variety of materials including :

- Stainless steel
- Hardened steel
- Cast iron
- Aluminum

The gears are suitable for reduction of high RPM electric motors for use in high torque low RPM applications. These gears are used in precision instruments because of their reliability and accuracy.

Use of Planetary Gears

These gears are the most widely used gears having diverse applications including :

- Sugar industry
- Power industry
- Wind turbines
- Marine industry
- Agriculture industry

Internal Gear



Internal Gear

An internal gear is a hollow gear with teeth cut on its internal surface. The teeth in such a gear project inwards instead of outwards from the rim.

Materials Used

There is a variety of materials being used to manufacture internal gears depending on the end application. Some of the popularly used materials are :

- Plastic
- Aluminum alloys
- Cast iron
- Stainless steel

The teeth in such gears can either be spur or helical. The internal teeth have a concave shape with a base thicker than that of an external gear. The convex shape and a strong base help in making the teeth stronger and also creating less noise.

Advantages of Internal Gear

- The gears are specially designed to accommodate a wide range of equipment.
- The gears are cost-effective and ideal for a broad range of light-duty applications.
- The non-binding tooth design ensures smooth and quiet operation.

Use of Internal Gears

- Light duty applications
- Rollers
- Indexing **External Gear**



External Gears

One of the simplest and most used gear units, external gears are extensively used in gear pumps and other industrial products for smooth functioning. These gears have straight teeth parallel to the axis. The teeth transmit rotary motion between parallel shafts.

Materials Used

The gears can be constructed from a variety of materials including:

- Stainless steel
- Hardened steel
- Cast iron
- Aluminum

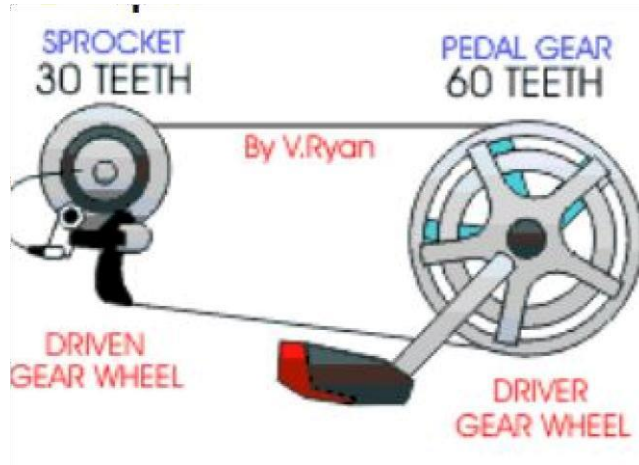
The kind of material used in manufacturing these gears depends on the end use they are being put to.

Use of External Gears

These gears are used in diverse fields including:

- Coal industry

- Mining
- Steel plants
- Paper and pulp industry



$$\frac{\text{Driven}}{\text{Driving}} = \frac{30}{60} = \frac{1}{2} \rightarrow 1:2$$

But WHAT does this mean? It means that the DRIVEN gear makes TWO rotations for every ONE rotation of the Driving Gear.

ASSIGNMENT

1. _____ can be defined as the force that resists the motion between two surfaces in contact or when one move relative to other.
 - A. mechanism
 - B. friction
 - C. fiction
 - D. friczing
 - E. metals

2. _____ is a mechanical energy transmission system of transmitting power from one sprocket wheel to the other on greased teeth

- A. chain drives
- B. belt drives
- C. friction
- D. electric grinder
- E. gear drive

3. _____ is a mechanical energy transmission system of transmitting power with the use of toothed edges wheel that mesh with one another

- A. gear drive
- B. chain drives
- C. belt drives
- D. friction
- E. electric movers

4. _____ Simply means the engagement of the teeth of the gears

- A. driven gear
- B. driving gear
- C. chain drives
- D. meshing of gears
- E. gear ratio

5. is the gear that transmits speed to the other gear

- A. driving gear
- B. gear ratio
- C. driven gear
- D. chain drives
- E. bevel gear

6. _____ is the gear that receives the impulse of the speed from the other gear

- A. gear ratio
- B. driven gear
- C. driving gear
- D. bevel gear
- E. chain drives

7. _____ can be defined as a relationship between two gears in which it shows how much larger the speed of one gear is than the other

- A. gear ratio
 - B. driven gear
 - C. driving gear
 - D. bevel gear
-

- E. chain drives
- 8. Describe the term "chain drives"
- 9. Describe the term "gears"
- 10. Highlight 5 types of gears

WEEK SIX: HYDRAULIC AND PNEUMATIC MACHINE

OBJECTIVES:

1. Identify the differences between Hydraulic and Pneumatics Devices.
2. Highlight all the purposes of Pneumatics devices.
3. Outline all the purposes of Hydraulic Devices
4. Describe the term 'Hydraulic and Pneumatics Devices'.

Pneumatics and Hydraulic Devices

Pneumatics is a device that makes use of compressed air to do useful work. Such devices make use of cylinders to produce force or linear motion at changing speeds. Generally, both pneumatics and hydraulic devices are used for the following purposes.

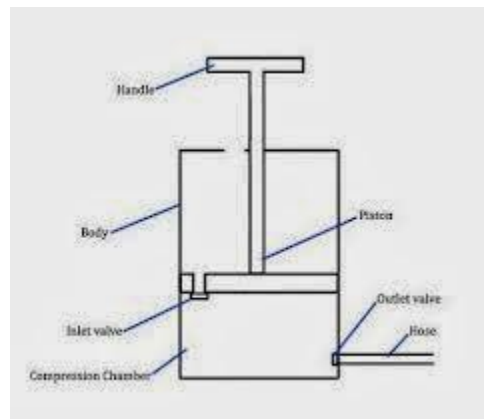
For pushing

For pulling and

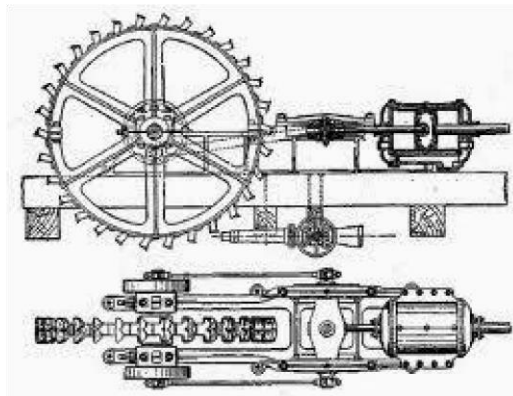
For lifting

Pneumatic Devices

Hand Pump: These are of different types, they are used to inflate tyres, balloons, balls etc. the hand pump has a plunger which sucks in and pushes out air at the barrel cylinder.



Compressor Machine: This is used by vulcanizers, while the modified one is used as a spray machine on several products.

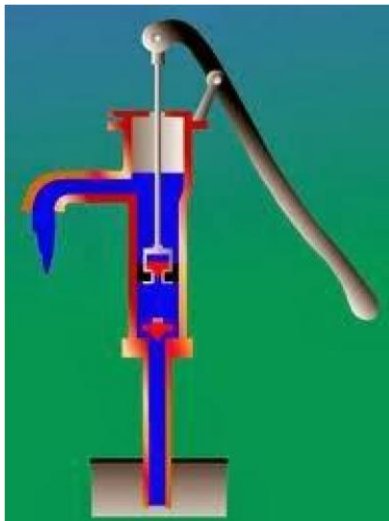


Hydraulic: These are devices that make use of liquids to push, pull and lift materials.

Hydraulic Devices

Hydraulic Jack: This jack has two main parts – the handle and the lever which is connected by two pistons which raises the lever that raises the load (car).

Suction and Double acting pumps: These are locally manually operated pumps used for lifting liquids and gases from reservoirs. e.g.

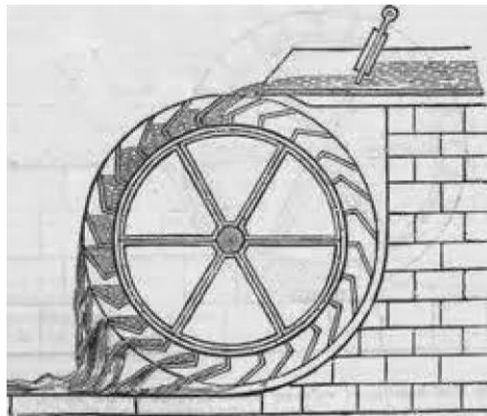


Garden Sprinkler: The sprinkler is used in water gardens. It has jets which issues out



water at a given speed about an axis.e.g.

Water Wheel: This wheel can be used to create mechanical energy in grinding, milling and electric power machines.



Others: (a) Reaction turbine (at dams) and (b) Brakes – used in cars to control motion.

ASSIGNMENT

1. ____ can be defined as the continuous and steady movement of air in a particular direction
 - A. air pump
 - B. air flow
 - C. human mouth
 - D. wind turbine
 - E. streamling
2. Airflow can be generated in the following ways except
 - A. fan
 - B. balloon
 - C. bellow
 - D. human mouth
 - E. air pump
3. ____ is a machine that grinds grain into flour
 - A. windmill

-
- B. streamlining
 - C. human mouth
 - D. wind turbine
 - E. wind power

4. _____ is the use of compressed air to make machine work.

- A. streamlining
- B. pneumatic
- C. wind turbine
- D. air pump
- E. windmill

5. _____ is an air pump that is driven by an electric motor or internal combustion engine

- A. compressor
- B. receiver
- C. air flow
- D. air pump
- E. wind turbine

6. The compressed air is usually stored in a strong metal tank called the

- A. receiver
- B. pump
- C. air pump
- D. turbine
- E. compressor

7. _____ is the flow of liquids or fluid through pipes under pressure in machineries for doing work.

- A. sliding doors
- B. hydraulic
- C. engine
- D. brake
- E. clutch

8. Highlight 2 devices that use pneumatics.

9. Outline 4 devices that make use of hydraulic.

10. Differentiate between pneumatics and hydraulic device.

WEEK SEVEN: MID TERM BREAK

WEEK EIGHT: SITE PREPARATION

OBJECTIVES:

1. Identify work done during site preparation.
2. Highlight all the uses of hand tools used in site preparation
3. Outline all the uses of mechanical tools used in site preparation
4. Describe the term 'site preparation'.

Definition:

Site preparation can be defined as the clearing and leveling of a piece of land for the purpose of building construction.

Hand Tools Used In Site Preparation

- (i) **Spade:** Spade is used for digging relatively loose or soft earth. Some persons do refer to the spade as 'shovel', which is technically wrong.
- (ii) **Shovel:** The shovel is used for lifting and throwing loose materials into another position, for spreading and leveling earth.
- (iii) **Machete:** The machete or cutlass is used for cutting grass, shrubs, wood-like trees and branches from sites.
- (iv) **Hoe:** Hoe is used in site preparation for digging the surface level of the ground.
- (v) **Axe:** The axe is used manually for cutting bigger branches of trees during site preparation. E.g.



Shovels A Spade A Machete



A Hoe Axes

Mechanical tools used in Site Preparation

- (i) **Chain saw:** This is a portable motorized saw used for felling trees and tree trunks into smaller sizes. E.g.



A Chain Saw

Bulldozer: This moving machine is used to push down any obstacle like big trees, old buildings and walls. E.g



- (ii) **Payloader:** The payloader is used for lifting large quantities of loose materials at a time (in whole) and loading it into trucks. E.g.



c

Payloader

- (iii) **Grader:** The grader is used for leveling excavated earth, trimming of edges of gutters and for cutting ditches. E.g.



A Grader at Work

Work Done During Site Preparation

- i. Removal of shrubs, grasses, trees and old buildings.
- ii. **Grubbing:** This is the act of removing roots of trees and stumps mechanically.
- iii. **Leveling:** There are two methods of leveling a piece of land to archive a suitable topography.
 - a. One method is called the cut and fill method i.e using the soil from a higher level of a piece of land to fill a lower part.
 - b. The other method is referred to as the earth-filling method where soil from another site is used to fill a swampy site before building construction takes place.
- iv. **Extermination** of termites and other insects.

ASSIGNMENT

- 1. Highlight and explain the uses five hand tools used in site preparation
- 2. Outline three work done during site preparation

WEEK NINE: SETTING-OUT

OBJECTIVES:

1. Highlight the materials used in setting-out.
2. Outline and explain the uses of the tools used in setting- out
3. Describe the term 'setting-out'.

Definition:

Setting-out can be defined as the process of driving wooden pegs into the ground in a way that tallies with the building plan or blue print. Lines are used to indicate boundaries on the dug wooden pegs during setting-out .

Materials Needed During Setting-Out

- (i) **Blue print or plan:** This is the drawing of the proposed building by an architect on paper. With the aid of the blue print, setting-out is carried out easily, since it indicates the dimensions of a building. E.g.



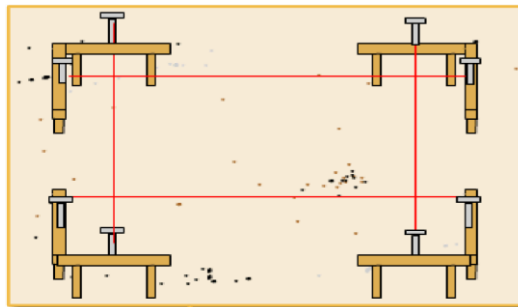
A Typical Blue Print

- (ii) **Wooden pegs:** These are shaped pieces of wood driven into the ground to run lines on. E.g.



Wooden Pegs

- (iii) **Wooden Profiles:** These are small sized plank used to mark the position of trenches and walls. E.g.



Wooden Profiles

- (iv) **Nails:** Nails are hammered on the pegs and profiles to run the lines. E.g.



Nails

- (v) **Lines:** These are nylon ropes used to indicate demarcation of the walls. E.g.



Tools Needed during Setting-Out

- (i) **Measuring tape:** The tape must be graduated in the metric form i.e millimeters, centimeters and meters. E.g.



A Measuring Tape

- (ii) **Sledge hammer:** This big hammer is used for driving pegs into the ground. E.g.



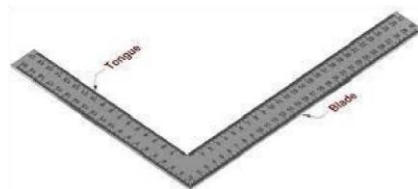
A Sledge Hammer

- (iii) **Builder's square:** This is a triangular wooden tool used for setting the squareness at the corners of the building space. E.g.



Builder's Square

- (iv) **Steel square:** The steel square serves the same purpose as the builder's square, just that it is made of a metallic material. E.g.



- (v) **Plumb:** The plumb is used to set walls in a vertical and horizontal position. E.g.



Spirit Level

Setting-out Terminologies

- (i) **Timbering:** This is the practice of using timber to support the vertical sides of a trench, so as to avoid collapse.
- (ii) **Excavation:** This is the digging up of earth to a required dept, width and length. This can be done both manually and mechanically.

ASSIGNMENT

1. _____ can be defined as an underground structure upon which a building is
 - A. erected
 - B. foundation
 - C. pillar
 - D. site plan
 - E. building plan
 - F. plan
2. The following are the types of foundation except
 - A. strip foundation
 - B. building foundation
 - C. pile foundation
 - D. raft foundation
 - E. stepped foundation
3. _____ is the digging of trenches for the building foundation using the foundation plan.
 - A. excavation

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- B. plan
 - C. soil
 - D. building
 - E. spade
4. The following are properties of organic soil except.
- A. it has fine particles
 - B. it is water retentive
 - C. it is soft
 - D. it is compressible
 - E. it is spongy and springy to walk on
5. Peat, humus and swampy soil can be regarded as ____ Soil
- A. fine grained
 - B. coarse grained
 - C. highly organic
 - D. clayed
 - E. sandy
6. Timbering depends on the following except
- A. depth of excavation
 - B. length of time
 - C. type of subsoil
 - D. climatic condition
 - E. tools
7. Choosing foundation depends on one of the following
- A. type of water to use
 - B. type of building
 - C. personnel
 - D. tools to use
 - E. cement to use
8. Describe the term setting-out
9. highlight at least four materials used in setting-out
10. outline and explain 4 uses of the tools used in setting- out
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